

KARAN SINGH

Faridabad, Haryana, India | +91- 7291010638 | karansingh06949@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFILE SUMMARY

Final-year Computer Science student and Full-Stack Developer with hands-on experience building and deploying enterprise-grade applications at CRIS (Ministry of Railways). Proficient in Angular, Node.js, LoopBack, PostgreSQL, REST APIs, and AWS. Passionate about writing clean, scalable code and solving complex engineering problems. Seeking a Software Engineer role where I can contribute to large-scale distributed systems.

SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, HTML5, CSS3

Frontend: Angular, RxJS, Angular Material, Responsive Design

Backend: Node.js, LoopBack 4, Express.js, REST APIs, GraphQL (basics)

Databases: PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS (EC2, S3, CloudFront, RDS), Git, GitHub, GitHub Actions, Docker (basics), Linux

Tools & Practices: VS Code, Postman, Swagger/OpenAPI, JWT, RBAC, Agile/Scrum

IoT & Messaging: MQTT, Apache Kafka, IoT protocols, real-time data streaming

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

CRIS – Centre for Railway Information Systems

New Delhi, India

Full Stack Developer Intern (Autonomous Society under Ministry of Indian Railways)

January 2026 – Present

- Assigned to the **RFID Group**, contributing to government-scale IT infrastructure powering Indian Railways' real-time asset tracking systems.
- Developed and maintained full-stack web features using **Angular** (frontend) and **Node.js + LoopBack** (backend) with PostgreSQL as the primary data store.
- Designed and integrated **RESTful APIs** consumed by multiple internal enterprise applications, improving inter-service data flow and reliability.
- Collaborated with senior engineers on optimizing database queries and backend services, reducing average response latency by identifying N+1 query patterns.
- Gained exposure to large-scale enterprise systems, government compliance standards, and production deployment workflows on **AWS** infrastructure.

PROJECTS

Persistent MQTT Broker with Kafka | Node.js, MQTT, Apache Kafka, PostgreSQL, Docker

- Built a custom **MQTT broker** from scratch in Node.js with full session persistence, ensuring retained messages and client subscriptions survive broker restarts.
- Integrated **Apache Kafka** as the messaging backbone to buffer and stream MQTT messages at scale, decoupling publishers from subscribers for high-throughput IoT workloads.
- Used **PostgreSQL** to persist session state, retained messages, and QoS packet queues; containerized the full stack with **Docker** and validated delivery guarantees under reconnection scenarios.

RFID Asset Tracking Dashboard | Angular, Node.js, LoopBack, PostgreSQL, REST APIs

- Built an internal dashboard for real-time RFID-based tracking of railway assets across stations, handling concurrent data streams from thousands of IoT endpoints.
- Implemented role-based access control (RBAC) and JWT authentication to secure sensitive government data.
- Wrote unit and integration tests reducing regression bugs in production by improving test coverage across critical API endpoints.

RESTful API Service with LoopBack 4 | Node.js, LoopBack 4, PostgreSQL, TypeScript

- Architected a modular REST API service with **LoopBack 4** and TypeScript, featuring auto-generated OpenAPI documentation, repository pattern, and dependency injection.
- Integrated **PostgreSQL** with complex relational schemas supporting many-to-many relationships and efficient indexed queries.
- Deployed the service on **AWS EC2** with environment-based configuration management and PM2 process management.

Cloud-Deployed Web Application | *Angular, AWS S3, CloudFront, EC2, Node.js*

- Deployed an Angular single-page application to **AWS S3** with **CloudFront** CDN, achieving sub-200ms global load times.
- Set up CI/CD pipeline using GitHub Actions for automated build, test, and deployment to AWS on every push to main branch.
- Configured **HTTPS**, custom domain routing, and caching policies, reducing bandwidth costs and improving page load performance.

EDUCATION

J.C. Bose University of Science and Technology, YMCA	Faridabad, India
Bachelor of Technology in Computer Science and Engineering CGPA: 7.0 / 10	2022 – 2026
Higher Secondary Education (Class 12) CBSE Board, Faridabad, Haryana (Percentage : 80%)	2021 – 2022
Senior Secondary Education (Class 10) CBSE Board, Faridabad, Haryana (Percentage : 88%)	2019 – 2020

ACHIEVEMENTS

- Government-Scale Deployment: Working on production systems serving Indian Railways, one of the world’s largest railway networks with over 1.4 billion annual passengers.
- Open Source: Actively contributing to personal and collaborative GitHub repositories; strong version control practices using Git workflows (branching, PRs, code review).
- Self-Driven Learning: Completed online coursework in AWS Cloud Fundamentals, Data Structures & Algorithms, and System Design through platforms like Coursera and LeetCode.